MILITARY SPECIFICATION

CLOTH, PARACHUTE, STRETCH AND RIPSTOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of nylon/spandex cloth, weighing 2.5 ounces per square yard.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-P-1133 - Packaging of Synthetic Fiber Fabrics.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFWAL/MLSE, ATTN: Standardization Manager, Wright-Patterson AFB OH 45433-6533 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-414 - Sampling Procedure and Table for Inspection by Variables for Percent Defective.
MIL-STD-851 - Coding: Manufacture's Color, Nylon Parachute Cloth.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be the one in effect on the date of the solicitation.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Application for copies should be addressed to the Federal Trade Commission, Washington DC 20580.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The nylon fiber used for the warp yarn in the manufacture of this parachute cloth shall be a bright, high tenacity, light and heat-resistant polyamide prepared from hexamethylene and adipic acid or its derivatives. The filling yarn shall be a mid-dull type. Both yarns shall have a melting point of 489 ±18°F (254 ±10°C) when tested in accordance with 4.3.2.4.

The spandex fiber used for the core of the filling yarn in the manufacture of this parachute cloth shall be a manufactured fiber in which the fiber-forming substance is a long-chain synthetic polymer comprised of at least 85 percent of a segmented polyurethane. It shall have a melting point of 446 ±18°F (230 ±10°C) when tested in accordance with 4.3.2.4. The yarns shall not be bleached in any manner or process.

3.1.1 Yarn make-up. The warp yarn shall be of 100 percent nylon fiber. The filling yarn shall have a core of 100 percent spandex (lycra) fiber and two wraps of 100 percent nylon fiber.

3.1.2 Fiber identification. Each roll of cloth shall be labeled or ticketed and invoiced for fiber content in accordance with the rules and regulations of the Textile Fiber Products Identification Act.

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3.1.3 Weave. The weave pattern shall be a plain one-up and one-down weave, except for the introduction of a ripstop weave in the fill yarn only. The ripstop shall be an 18 pick repeat with two double picks. See figure 1 for situation woven as one pick.

3.2 Physical properties. The finished cloth shall conform to table I.

TABLE I. Physical properties.

Weight, ounces per square yard (max.)		2.5
Thickness, inches (max.)		0.010
Breaking strength, pounds per inch (min.) (ravel strip)	Warp	60
	Filling	40
Elongation, percent (min.)	Warp	35
	Filling	85
Tear strength, pounds (min.) (tongue)	Warp	2.8
	Filling	2.8
Yarns per inch	Warp	186 ±5
	Filling	145 ±4
Twist, Z turns per inch (min.)	Warp	7.5 Z
	Filling - based on amount of double cover fibers to give an elongation of 160 percent	
Yarn denier	Warp	30
	Filling - 40 denier spandex (lycra) core - double covers opposing directions of 15 denier nylon	

Air permeability - cubic feet of air per minute per square foot of cloth

	Water pressure	Min.	Max.
Test 1 1/	at 0.5 in.	70	110
	at 6.0 in.	500	700
	at 12.0 in.	850	1150

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TABLE I. Physical properties - Continued.

Air permeability - cubic feet of air per minute per square foot of cloth				
Test 2 2/	Loading per leg of cruciform test specimen (lbs./in.)	Water pressure at 0.5 in. min.	Water pressure at 0.5 in. max.	Water pressure at 2.0 in. min.	Water pressure at 2.0 in. max.
	1	160	220	440	560
	2	205	265	540	660
	3	235	295	580	700
	4	255	315	610	730
	5	275	335	640	760

1/ See figure 2 for graphical representation of this requirement.

2/ Permeability/pressure graphs (see figure 3) represent these requirements. Also see 4.4.2.1.1.

NOTE: max. - maximum
min. - minimum

3.3 Dimensions.

3.3.1 Width. The overall width of the finished cloth shall be 41.5 ±.5 inches (see 6.2), after 24 hours of relaxation under ambient conditions.

3.3.2 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be in continuous lengths of not less than 100 yards. Shorter cuts may be included in accordance with the following schedule:

75 percent of total yardage in cuts 100 to 250 yards

15 percent of total yardage in cuts 25 to 100 yards

10 percent of total yardage in cuts 15 to 25 yards.

The pieces from one finisher's roll shall be put up in rolls as specified in 5.1.

3.4 Finish and color. The cloth shall be given a preliminary scour sufficient to remove sizing and other contamination, at a temperature which will not cause fixation of same into the cloth or result in permanent setting of the fabric. The cloth shall be dried and heat-set at sufficient temperature so as to control the air permeability in the finished cloth.

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The cloth shall not be calendered, and shall not be bleached in any manner of process. The color shall be natural.

3.4.1 Permanence of finish. The permanence of the cloth finish shall be such that when the cloth is subjected to the test specified in 4.4.2.1:

a. The average of the air permeability readings taken after testing shall be within 20 percent of the average of the readings taken before testing.

b. The cloth thickness after testing shall not exceed 15 percent more than the thickness before testing.

c. The cloth shrinkage measured after testing shall not exceed 2 percent in either the warp or filling direction.

3.4.2 Stability of finish. The stability of the cloth finish shall be such that when the cloth is tested as specified in 4.4.2.2, the change in air permeability shall not be greater than 7.5 cubic feet of air per minute per square foot of cloth.

3.5 Permeability change after accelerated aging. The change in the permeability, as defined in table I, shall be within the limits allowed by 3.4.1a after accelerated aging per method 5850, FED-STD-191, 48 hours at 212 to 221°F (100 to 105°C). Test specimen size for biaxial tension testing shall be as shown in figure 4.

3.6 Non-fibrous materials. The chloroform soluble material of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.4.

3.7 Resistance to light and heat. The finished cloth shall not lose more than 25 percent warpwise and 45 percent fillingwise of its original strength when light tested and shall not lose more than 25 percent of its original strength when heat tested in accordance with 4.4.3.

3.8 Acidity-alkalinity (pH). The pH value of the finished cloth shall be within the range of 5.0 to 8.5 when tested as specified in 4.4.

3.9 Identification and marking.

3.9.1 Manufacturer's identification. Colored threads shall be woven into the central area of the cloth to identify the manufacturer of the cloth to the procuring activity. The colors used shall be those indicated in MIL-STD-851. The dye used in the marker threads shall have no deleterious effect on the physical properties of the thread, and the dyed thread shall show no bleeding or color transfer when subjected to the tests for laundering and dry cleaning as specified in 4.4.

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3.9.2 Identification of product. Each roll of finished cloth shall be marked for identification in accordance with PPP-P-1133. In addition, each piece of cloth in each roll shall be clearly and legibly marked with the finisher's roll number or code, as specified in 4.3.1a, and each roll shall have attached a durable tag on which the finisher's roll number or code is listed. The date of manufacture of the cloth shall be included on the tag attached to each roll.

3.10 Age. The yarn shall be not more than three years old from the date of manufacture to the date of delivery of the cloth.

3.11 Workmanship. The finished cloth shall be clean and evenly woven and shall conform to the quality and grade of product established by this specification. The finished cloth shall be free from defects except to the extent specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The examination and testing of nylon/spandex (lycra) parachute cloth shall be classified as quality conformance inspection.

4.3 Quality conformance inspection. Quality conformance inspection shall be conducted at a laboratory satisfactory to the procuring activity. Inspection shall consist of the examination and tests specified in table II. In addition to test method paragraphs, reference is made to methods in FED-STD-191.

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TABLE II. Quality conformance inspection.

Test characteristics	Requirement paragraph	Test method
Weave	3.1.3	Visual
Weight	Table I	5041
Thickness	Table I	5030
Breaking strength and elongation (ravel strip) 1/	Table I	5104
Tearing strength (tongue)	Table I	5134
Yarns per inch	Table I	5050
Air permeability	Table I	4.4.1
Width	3.3.1	5020
Permanence of finish thickness	3.4.1	5030 & 4.4.2.1
Permanence of finish shrinkage	3.4.1	4.4.2.1
Stability of finish	3.4.2	4.4.1 & 4.4.2.2
Chloroform soluble matter	3.6	2611
Resistance to light	3.7	4.4.3.1
Resistance to heat	3.7	4.4.3.2
Acidity-alkalinity (pH)	3.8	2811

1/ Except that the jaws of the clamp shall be 1 inch by 1.5 inches or more with the long dimension of the jaws perpendicular to the application of the load.

4.3.1 Inspection term definitions. The tests and examination shall consist of the following:

a. A finisher's roll is the roll formed when lengths of cloth are seamed together for the purpose of finishing. Each finisher's roll shall be marked or tagged with a number or code for identification purposes. Each finisher's roll shall be tested in accordance with sampling plan A (see 4.3.2.1).

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b. Shipper's rolls are the rolls created after finishing when the finisher's roll is divided for the purpose of handling and delivery. Each shipper's roll shall be tagged with the number or code of the finisher's roll of which it was a part. Shipper's rolls shall be tested and examined in accordance with sampling plans B and C (see 4.3.2.2 and 4.3.2.3).

c. Pieces are the unspliced, unseamed, lengths of cloth. Each piece shall be marked with the number or code of the finisher's roll from which it was formed. Each piece shall be examined in accordance with 4.3.2.4.

MIL-STD-109 shall apply for the inspection terms used herein.

4.3.2 Sampling plans-cloth.

4.3.2.1 Sampling plan A and tests.

4.3.2.1.1 Lot. A lot shall consist of each finisher's roll (see 4.3.1).

4.3.2.1.2 Sampling. Each sample shall be of sufficient length and full width of the cloth to perform tests required. The number of samples required (sample size) shall be determined in accordance with inspection level III in table titled "Sample size code letters" of MIL-STD-414. The distance between samples shall be determined by dividing the number of samples required into the lot size. The first sample shall be selected at the beginning of the lot and the succeeding sample selected by advancing into the lot the distance required between samples. This shall be repeated until the required number of samples have been selected. The actual distance between samples may vary from the calculated distance by ±40 percent. This latitude is permitted so that samples may be selected near an existing seam thereby reducing the number of short pieces. No sample, however, shall be selected within 28 inches of a seam. Exception to the above sampling is authorized when total lot size of material being procured is 1,000 yards or less. On these small lots, one test for air permeability will be taken on each end of every piece of cloth comprising the lot.

4.3.2.1.3 Test and acceptance criteria. Each sample selected in accordance with 4.3.2.1.2 shall be subjected to the air permeability tests as specified in 4.4.1. The acceptable quality level shall be 2.5 percent. Acceptance of the lot shall be determined in accordance with the standard deviation method, double specification limit, variability unknown of MIL-STD-414. On lots less than 1,000 yards, any test failure will cause rejection of the piece of cloth represented.

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4.3.2.2 Sampling plan B and tests.

4.3.2.2.1 Lot. A lot shall consist of each shipper's roll formed from one finisher's roll (see 4.3.1).

4.3.2.2.2 Sampling. One sample at least 1 yard in length, the full width of the cloth and representative of the roll, shall be selected from each shipper's roll and subjected to the tests specified in 4.4. Failure of the sample to conform to the following tests shall reject the lot represented. (See table II).

a. Weight

b. Thickness

c. Breaking strength and elongation

d. Tearing strength

e. Yarns per inch

f. Air permeability (see table I)

4.3.2.3 Sampling plan C and tests.

4.3.2.3.1 Lot. A lot shall consist of all shipper's rolls formed from one finisher's roll (see 4.3.1).

4.3.2.3.2 Sampling. A 4-yard, full-width sample shall be selected and subjected to the tests specified in 4.4. Failure of the sample to conform to the following tests shall reject the lot represented.

a. Weave

b. Permanence of finish

c. Stability of finish

d. Chloroform soluble material

e. Light and heat resistance (see 3.7)

f. Acidity-alkalinity (pH)

g. Colorfastness to light, dry cleaning, laundering water

4.3.2.4 Sampling plan D and testing of components. In addition to the quality assurance provisions of the subsidiary specifications, components and materials listed in table III shall be tested for the

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characteristics specified and in accordance with the referenced test methods of FED-STD-191. The lot size shall be expressed in units of one cone, tube, or spool each. The unit of product shall be one cone, tube, or spool for each component. The sample size shall be in accordance with special inspection level S-1 in table I of MIL-STD-105. The acceptance quality level for each characteristic shall be 6.5 test failures per 100 units.

TABLE III. Components and materials.

Material	Characteristics	Spec. requirements	Test method	No. determinations per individual unit of product	Results reported
Warp and filling	Tenacity	3.1	1/	see 4.4.3	Pass or fail
Yarns	Luster	3.1	1/	5	Pass or fail
	Melting point	3.1	1534	2	Average of 2 determinations to nearest degree centigrade
	Bleaching	3.1	1/	see 4.4.3	Pass or fail
	Light and heat resistance	3.1	see 4.4.3	1	Pass or fail
	Twist per inch	table I	4052	5	Average of 5 determinations to nearest 0.5 turns per inch
Identification yarns	Deleterious effect of dye	3.9.1	1/	1	Pass or fail
	Colorfastness to laundering	3.9.1	5614	1	Pass or fail on bleeding and color transfer
	Dry cleaning	3.9.1	5620	1	Pass of fail

1/ Acceptance with respect to this characteristic will be based on a contractor's certification of compliance.

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4.3.2.5 Sampling plan E for examinations.

4.3.2.5.1 Lot. A lot shall consist of a shipper's roll from each finisher's roll (see 4.3.1).

4.3.2.5.2 Yard-by-yard examination. Each shipper's roll shall be visually examined yard-by-yard for defects as specified in 4.3.3. No shipper's roll shall contain more than 7 major defects over 100 linear yards.

4.3.2.5.3 Overall examination. Each shipper's roll shall be visually examined for overall defects as specified in 4.3.4.

4.3.2.5.4 Visual examination criteria. A shipper's roll shall contain no defects as specified in 4.3.4.

4.3.3 Yard-by-yard examination. All cloth shall be carefully examined yard by yard and the defects classified in accordance with table IV. The unit of product for this examination shall be one linear yard (that is, the increment of one yard on the measuring device of the inspection machine).

TABLE IV. Classification of defects.

Defect	Description	Major	Minor
Abrasion	Any abrasion mark showing fussiness	X	
Biased filling	Biased filling more than 2 inches from horizontal at greatest point of bias	X	
Bowed filling	Filling bow more than 2 inches in height (as measured from a straight line cord to highest point of arc)	X	
Broken or missing end	Two or more contiguous, regardless of length	X	
	Single, more than 36 inches long	X	
	Single, 36 inches long or less		X
Broken or missing pick	Two or more contiguous, regardless of length	X	
	One pick, full width		X
Cut, hole, or tear (other than pinholes)	Three or more warp of filling threads ruptured at adjoining points	X	

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TABLE IV. Classification of defects - Continued.

Defect	Description	Major	Minor
Floats or skips	Any multiple float 3/16 inch square or more	X	
	Single floats 1/4 inch or more in length	X	
	Contiguous floats or pin floats 1/ the sequence of which measure 1 inch or more in length	X	
	Any multiple float up to 3/16 inch square		X
	Single floats up to 1/4 inch in length		X
	Contiguous floats to pin floats 1/ the sequence of which measures less than 1 inch in length	X	
Heavy filling bar or heavy place	Over 1/8 inch in width and varying 10 percent or more from normal pick count	X	
	Over 1/2 inch in width and varying less than 10 percent from normal pick count	X	
	1/8 inch or less in width and varying 10 percent or more from normal pick count		X
	1/2 inch or less in width and varying less than 10 percent from normal pick count		X
Jerked-in filling	Any jerked-in filling occurring more than 4 times within 10 linear inches	X	
	Any jerked-in filling occurring 4 times or less within 10 linear inches		X
Loops, kinks, or snarls	All over 1/8 inch in length	X	
	Three or more (in any linear yard) up to 1/8 inch in length		X
	Up to two (in any linear yard) up to 1/8 inch in length		X

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TABLE IV. Classification of defects - Continued.

Defect	Description	Major	Minor
Mispick or double pick	Three or more additional picks in the shed	X	
	Two picks		X
Pinholes or yarn deformations	Over 6 pinholes or yarn deformations occurring within an area equal to a 6-inch diameter circle	X	
	Three to six pinholes or yarn deformations occurring within an area equal to a 6-inch diameter circle		X
Spot, stain, or streak	Single ends or picks 15 inches or more in length	X	
	Double ends or picks 8 inches or more in length	X	
	Over 2 ends or picks 5 inches or more in length or a clearly noticeable area more than 1/4 square inch in area, whichever is greater	X	
	Single ends or picks 2½ inches up to 15 inches in length		X
	Double ends or picks 2½ inches up to 8 inches in length		X
	Over 2 ends or picks less than 5 inches in length or a clearly noticeable area 1/4 square inch or less in area, whichever is greater		X
Slubs or strip back 2/	More than 5 over 1/4 inch in length	X	
	Two up to and including 5 over 1/2 inch in length	X	
	One over 1 inch in length	X	
	Five or less over 1/4 inch but not exceeding 1/2 inch in length		X

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TABLE IV. Classification of defects - Continued.

Defect	Description	Major	Minor
	One over 1/2 inch but not exceeding 1 inch in length		X
Weave	Pattern not conforming to weave diagram specified	X	
Width	Less than specified	X	
Wrong draw	Resulting in clearly noticeable warpwise streak more than 18 inches in length	X	
Manufacturer's (weaver) identification yarn	Missing	X	

1/ A pin float is defined as a float measuring 1/8 inch or less. Single pin floats shall not be considered as defects.

2/ A strip back is defined as a broken filament(s) wrapped around the remaining yarn forming an enlarged area resembling a slub.

4.3.3.1 Calculating the defects. For each unit of product the defects shall be counted as follows:

a. One minor defect and one major defect shall be counted as one major defect.

b. Three or more minor defects shall be counted as one major defect.

c. One or more major defects shall be counted as one major defect.

d. A continuous major defect shall be counted as one defect for each unit of product or fraction thereof in which it occurs.

4.3.3.2 Flagging the defects. Each major defect shall be flagged by a red string sewn in the selvage. Three or more minor defects occurring in a yard shall be flagged by a red string in the selvage. A continuous major defect shall be flagged by a single yellow string sewn into the selvage for each yard containing the defect.

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4.3.3.2.1 Allowance for defects. A one-half yard allowance shall be made for each major defect flagged except for continuous defects which shall be given a one-yard allowance for each yard in which the defect occurs.

4.3.3.3 Fiber identification. During the yard-by-yard examination each roll shall be examined for fiber identification. The lot shall be unacceptable if two or more rolls in the sample are not labeled in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

4.3.3.4 Examination for length.

4.3.3.4.1 Individual rolls. During the yard-by-yard examination, each roll shall be examined for length. Any roll length found to be less than the minimum or maximum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length.

4.3.3.4.2 Total yardage. The lot shall be unacceptable if the total of the actual lengths of rolls examined is less than the total of the length marked on the ticket.

4.3.4 Overall examination. The unit of product for overall examination shall be one piece (see 4.3.1). Each piece shall be carefully examined and shall contain none of the following defects:

a. Objectionable odor

b. Uncleanliness throughout

c. Uneven weaving throughout.

4.4 Test methods. The methods of testing specified in FED-STD-191, wherever applicable, as listed in table II, and as specified in 4.4.1 through 4.4.4 shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on the applicable unit of product noted for test purposes as specified in the applicable test methods.

4.4.1 Air permeability. The test specimen for table I shall be 7 inches long and the full width of the cloth. The air permeability test shall consist of five individual readings made in accordance with method 5450 of FED-STD-191. The individual readings shall be equally spaced across the width (between selvages) of the test specimen except that no readings shall be taken within an area from the selvage equal to 10 percent of the specimen width. The air permeability of the test specimen shall be the arithmetic mean or average of the five individual readings. Test 2 in table I shall be as specified in 4.4.2.1.1.

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4.4.2 Finish.

4.4.2.1 Permanence of finish. Two 20-inch square specimens of the cloth shall be prepared. Using a template and indelible ink, an 18-inch square shall be marked on each specimen. The specimens shall be subjected to the air permeability and thickness tests in accordance with table I and method 5030 of FED-STD-191 respectively. A container of adequate size to accommodate both specimens, prepared as described below, shall be filled to within 3 inches of the top with water which shall be heated to a rapid boil. Both specimens shall be placed in the boiling water in a skein-type form prepared by stapling the two opposite sides of a specimen together to form a loop or skein. One specimen shall have the warp yarns vertical in the skein and the other specimen shall have the filling yarns vertical. Each specimen shall then be placed over a glass rod 1/4 inch in diameter and 21 inches in length. A glass tube or rod 1/4 inch in diameter, 21 inches in length, and approximately 45 ±5 grams in weight shall be placed inside each loop at the bottom. Both loops shall then be suspended freely in the boiling water bath by attaching each with twine or wire to glass rods which are 1/4 inch in diameter and of sufficient length to rest on the top of the container. The specimen shall be subjected to the action of the boiling water bath for a period of 15 minutes, after which they shall be removed from the bath and allowed to drain for a few minutes. The staples shall be removed from the specimens, and the specimens shall be placed flat on a horizontal screen to air dry. After the specimens are thoroughly dry they shall be exposed for at least 4 hours to a standard atmosphere of 65 ±2 percent relative humidity and a temperature of 70 ±2°F (21.1 ±1.1°C). The 18-inch square shall be measured to the nearest 0.01 inch in 6 places, 3 in the warp direction and 3 in the filling direction. The results of the warp skein and filling skein for thickness, air permeability, and shrinkage shall be averaged, and acceptance shall be based on the average results. The percentage of shrinkage in either the warp or filling direction shall be computed as follows:

$$\frac{\text{18 in. - distance between marks after boiling}}{\text{18 in.}} \times 100 = \text{percent shrinkage}$$

The specimen shall again be subjected to the air permeability and thickness test to determine conformance to 3.4.1.

4.4.2.1.1 Preparation of special test specimens for biaxial load permeability tests. In addition to the 20-inch square specimens (see 4.4.2.1), other specimens 32-inch square shall also be prepared. Two specimens shall be used for preparation of cross-shaped biaxial tension test samples, as shown in table V. Two other square specimens will be subjected to treatment as specified in 4.4.2.1, as applicable; mark 30-inch squares in indelible ink in lieu of 18-inch squares. The percentage of shrinkage in either the warp or filling direction shall be computed as follows:

$$\frac{\text{30 in. - distance between marks after boiling}}{\text{30 in.}} \times 100 = \text{percent shrinkage}$$

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TABLE V. Permeability versus biaxial load.

Loading per leg of cruciform specimen (lbs./in.)	Pound weights to use on each strap
1	5
2	10
3	15
4	20
5	25

Flow measurements shall start at the lowest tension of 1 pound per inch, and tension shall be increased in increments of 1 pound per inch up to 5 pounds per inch. This loading shall be applied over an area that can be simultaneously installed as test piece on the Frazier permeability tester. (See figures 5 and 6 and paragraph 6.3.) When applying loads, the point loads on any of the edges of the test specimen shall be avoided, and all tensions shall be evenly distributed over any linear edge. After recording the loading results, the tension shall be reduced in increments of 1 pound per inch down to 1 pound per inch and the results recorded.

4.4.2.2 Stability of finish. The specimen shall consist of a 24-inch square of cloth measured to within an accuracy of 1/2 inch. Five circles shall be marked on the specimen. The circles shall have a diameter equal to but not exceeding 2 inches more than the diameter of the cloth orifice on the test equipment. One circle shall be positioned at each of the four corners on the diagonal line connecting opposite corners and at 1/3 of the distance of the diagonal center of the specimen. A single air permeability reading shall be made within each circled area in accordance with method 5450 of FED-STD-191. One corner of the specimen shall be inserted through a bushing conforming to figure 7 and the specimen pulled through the bushing and back again for 2½ cycles for a total of 5 passes through the bushing. One of the corners adjacent to the first corner shall then be inserted through the bushing and the procedure repeated until the specimen has been pulled through the bushing for 5 complete cycles (a total of 10 passes through the bushing). The specimen shall be pulled straight through the bushing by pulling the specimen in the vertical direction. Care shall be taken to insure that yarns that might ravel out the specimen do not become tangled with the bushing and restrain the specimen when being pulled through the bushing. The specimen shall again be subjected to the air permeability test. The average of the initial readings shall be compared with the average of the final readings to determine conformance to 3.4.2 and table I at 0.5 inch water pressure.

4.4.3 Resistance to light and heat. The test specimens for determining resistance to light and heat shall be so selected that the identical warp and filling yarns are tested originally and after subjection to light

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and heat. This shall be done by marking all the specimens required with dye resistance ink before cutting. The specimens shall be so cut and marked that specimens in each group--that is, warp and filling--be tested for original breaking strength, for resistance to light and resistance to heat containing identical yarns. (See figure 5.)

4.4.3.1 Resistance to light. Specimens of fabric (see figure 8) shall be exposed in the accelerated weathering unit as specified in test method 5804 of FED-STD-191. A sufficient quantity of specimens to make five warp and five filling breakage tests shall be exposed. Corex D filters and Sunshine carbons shall be used. The total exposure time shall be 50 hours at 55 percent ±5 relative humidity and black panel temperature of 155 ±10°F (68.3 ±5.6°C). The spray heads shall be shut off during the entire exposure period. At the start of each exposure cycle (approximately 17 hours), the drainage pan shall contain water to a depth of approximately 3/4 of an inch. At the end of the exposure period the specimens shall be brought to equilibrium under standard conditions and shall then be tested for breaking strength as specified in table I, and the percent of breaking strength lost shall be calculated as follows:

$$\frac{\text{Original breaking strength - breaking strength after aging}}{\text{original breaking strength}} \times 100 = \text{percent of breaking strength lost}$$

4.4.3.2 Resistance to heat. Five warp and five filling tests shall be made. The test specimens shall be suspended in a circulating air oven at a temperature of 198 ±2°F (92 ±1°C) for one hour. (See figures 9 and 10.) After removal from the oven, the specimens shall be brought to equilibrium under standard conditions as defined in FED-STD-191. They shall then be tested for breaking strength as specified in table I, and the percent of breaking strength lost shall be calculated as in 4.4.3.1.

4.4.4. Permeability resistance to heat.

4.4.4.1 Test 1. Three test specimens, full width of cloth and 7 inches long, shall be tested for permeability according to table I (test 1). After accelerated aging per method 5850, FED-STD-191, 48 hours at 212 to 221°F (100 to 105°C), followed by 16 to 96 hours at standard atmospheric conditions, the specimens shall be tested again for the same properties.

4.4.4.2 Test 2. Six test specimens conforming to figure 4 shall be prepared, taken from a single piece of cloth adjacent to each other. Three specimens shall be tested immediately for permeability according to table I (test 2) biaxial loading. Three other specimens shall be subjected to accelerated aging as in 4.4.4.1 and then tested for the permeability according to table I (test 2).

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4.5 Examination and preparation for delivery. An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking requirements of section 5 of this specification are complied with.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Levels A and commercial. The cloth, put up as specified, shall be packaged in accordance with PPP-P-1133.

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

5.2.1 Levels A, B or commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

6. NOTES

6.1 Intended use. The primary use of this cloth is in the manufacture of emergency escape personnel parachutes.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification

b. Type (if applicable)

c. Color (see 3.4)

d. Quantity

e. Width, if other than specified in 3.3.1

f. Whether short lengths are acceptable (see 3.3.2)

g. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Frazier permeability tester device. The apparatus used at Irvin Industries Canada, Ltd. of Fort Erie, Canada provides a descriptive example of the test method for determining permeability versus biaxial load curve.

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The device consists of a slotted board installed around the Frazier permeability tester. The test piece is in the shape of a cross with a 5-inch by 5-inch center area that fits over the flow orifice. The loading is applied by hanging weights on the ends of the cross-shaped test piece that pass over the rollers and through slots. After applying gravity loading, the center of the test piece is clamped into position and is ready for the permeability test.

Custodians:
Army - GL
Navy - AS
Air Force - 20

Reviewer activities:
Army - MD
Navy - SH
Air Force - 99

User activity:
Navy - OS

Preparing activity:
Air Force - 20

(Project 8305-0913)

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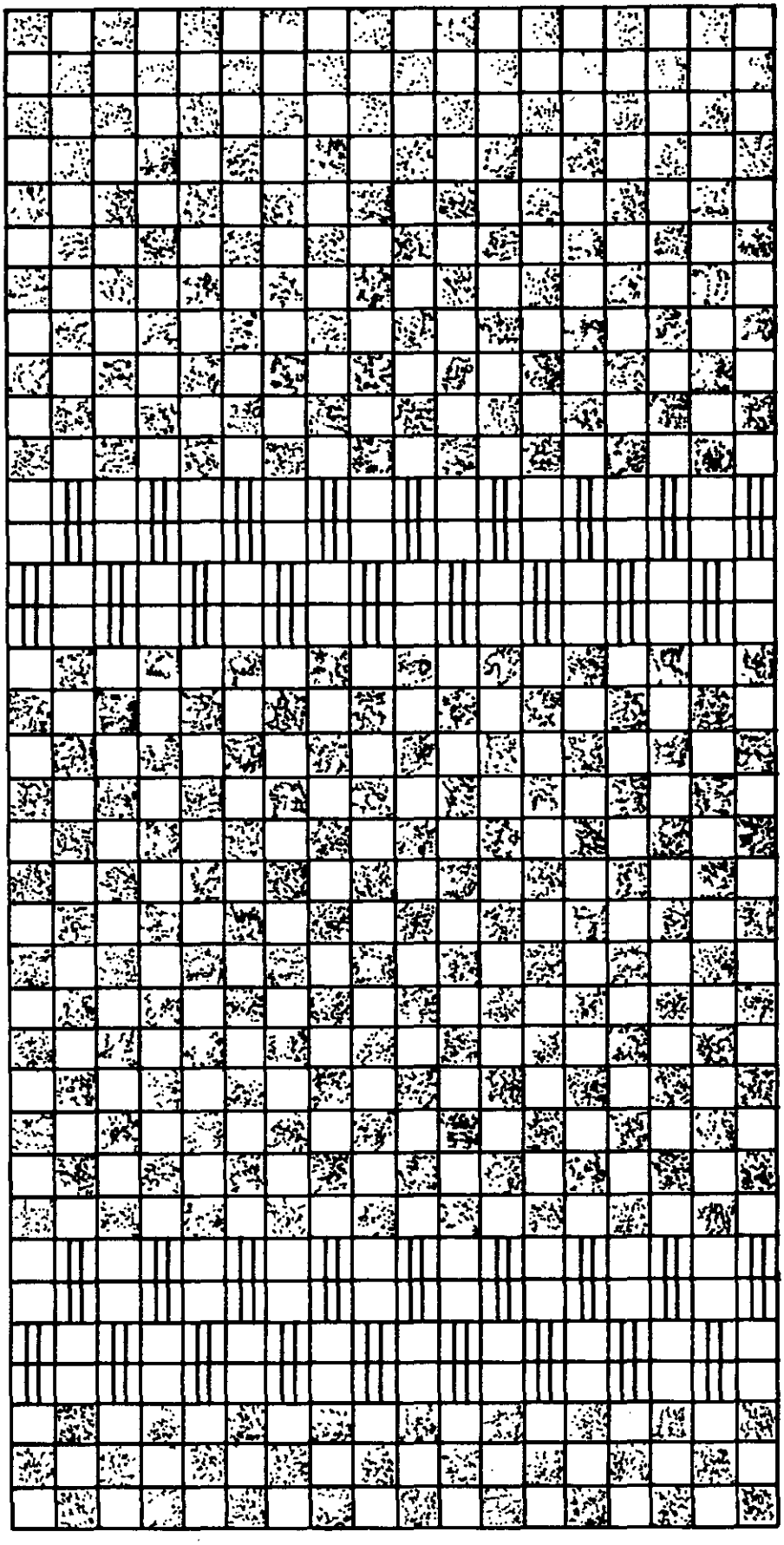


FIGURE 1. Weave pattern.

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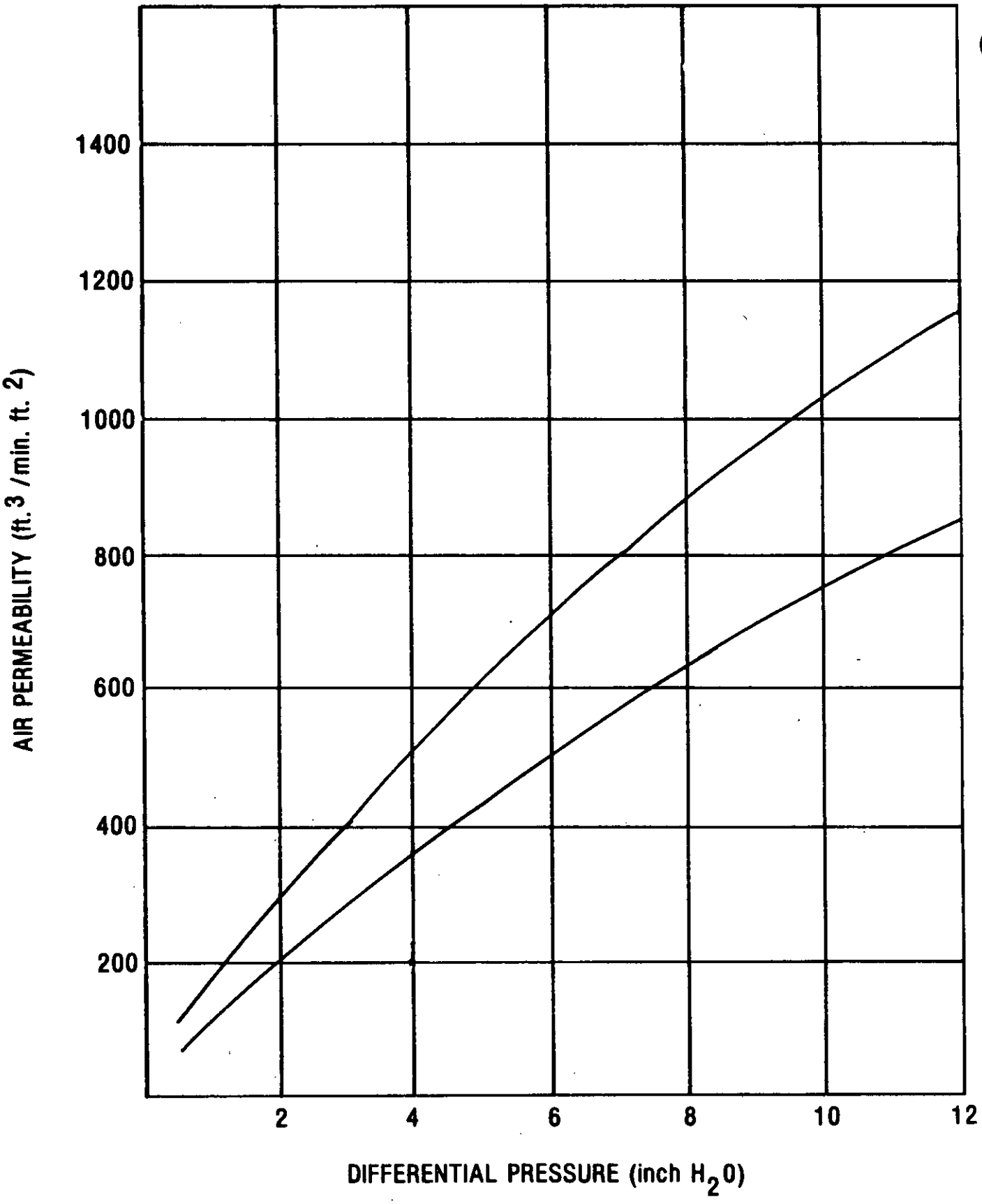


FIGURE 2. Air permeability - test 1.

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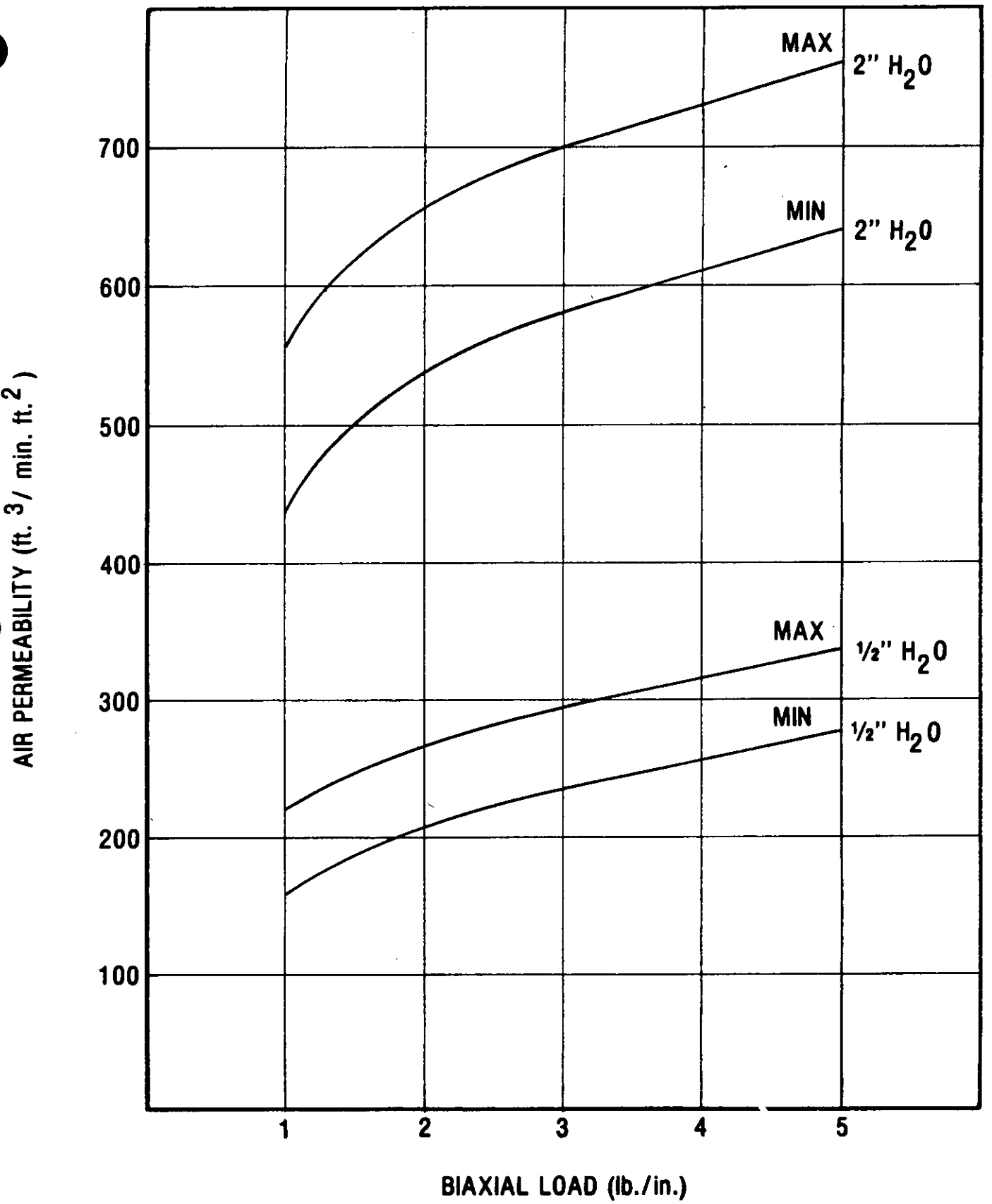


FIGURE 3. Air permeability - test 2.

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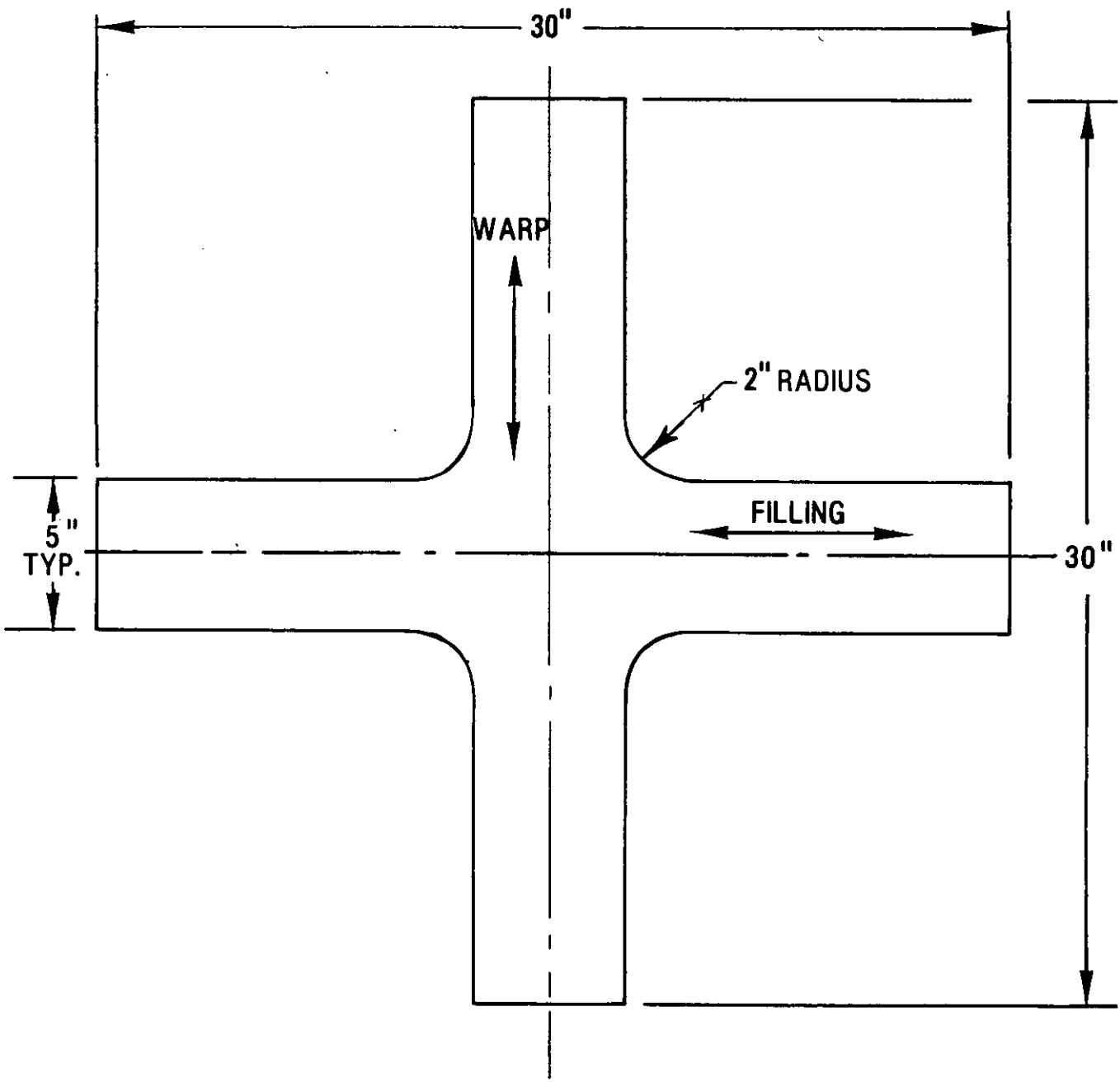


FIGURE 4. Stretch fabric Frazier air permeability template (biaxial load).

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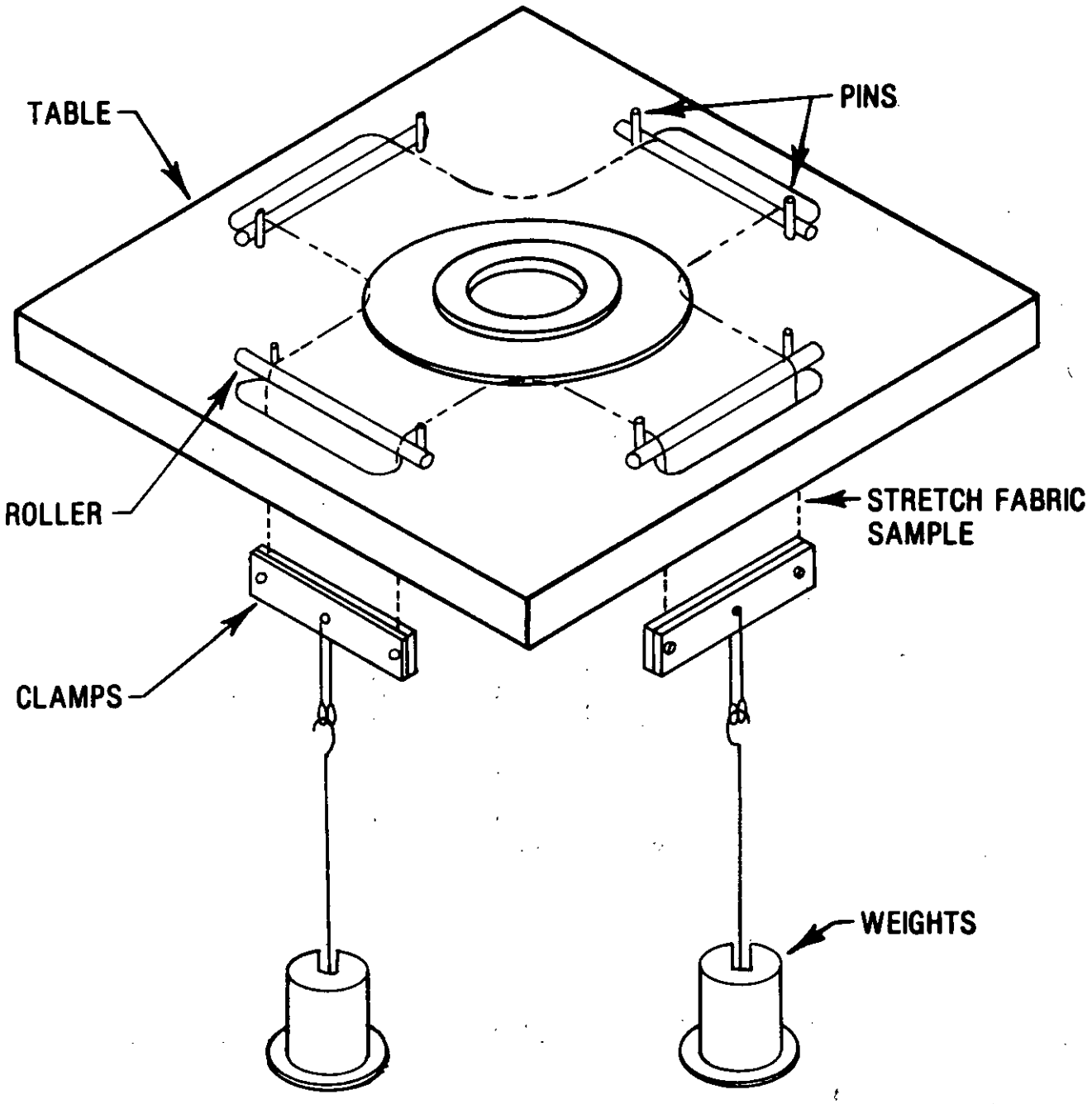


FIGURE 5. Attachments to Frazier air permeability test machine (biaxial load).

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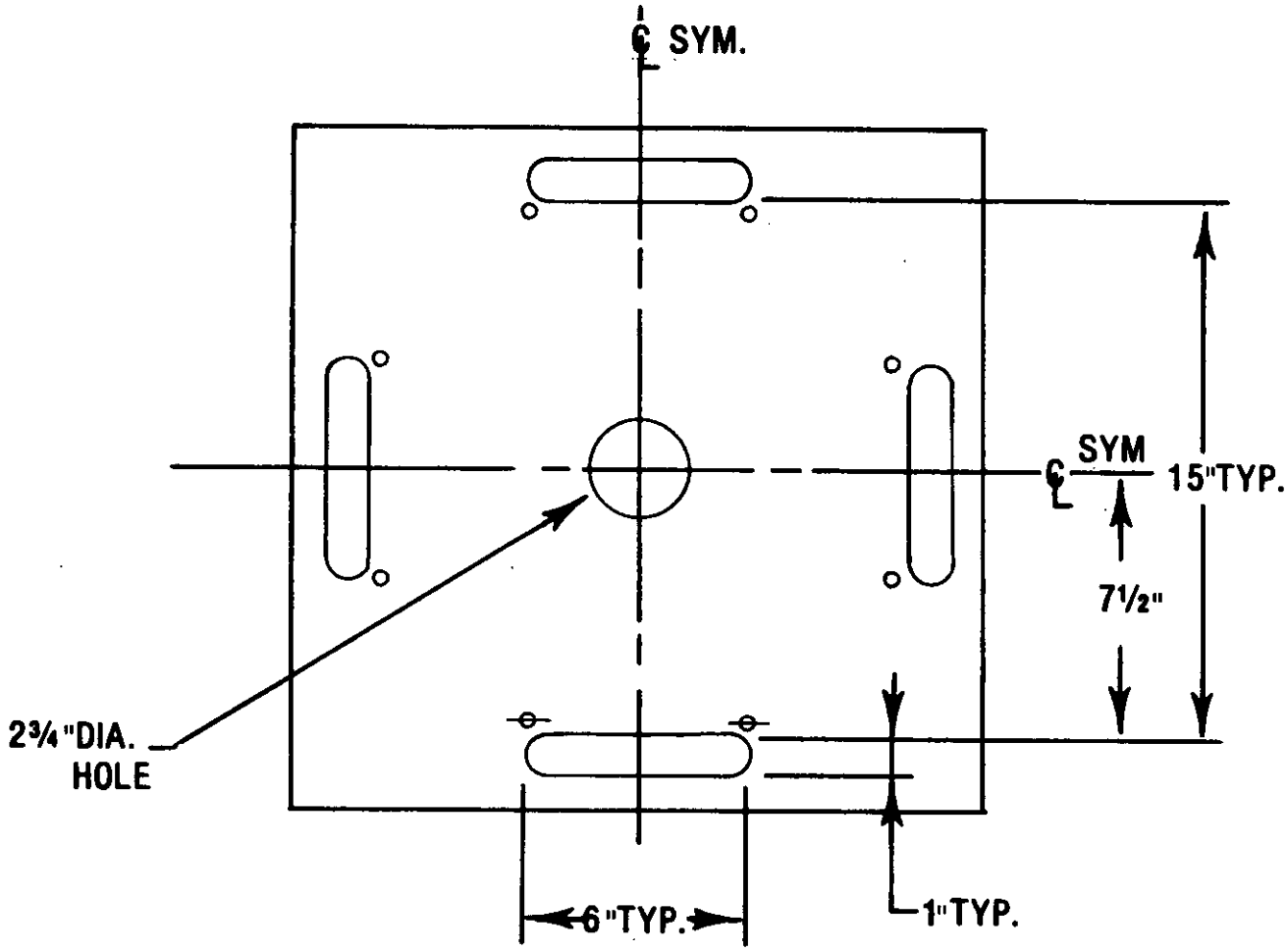
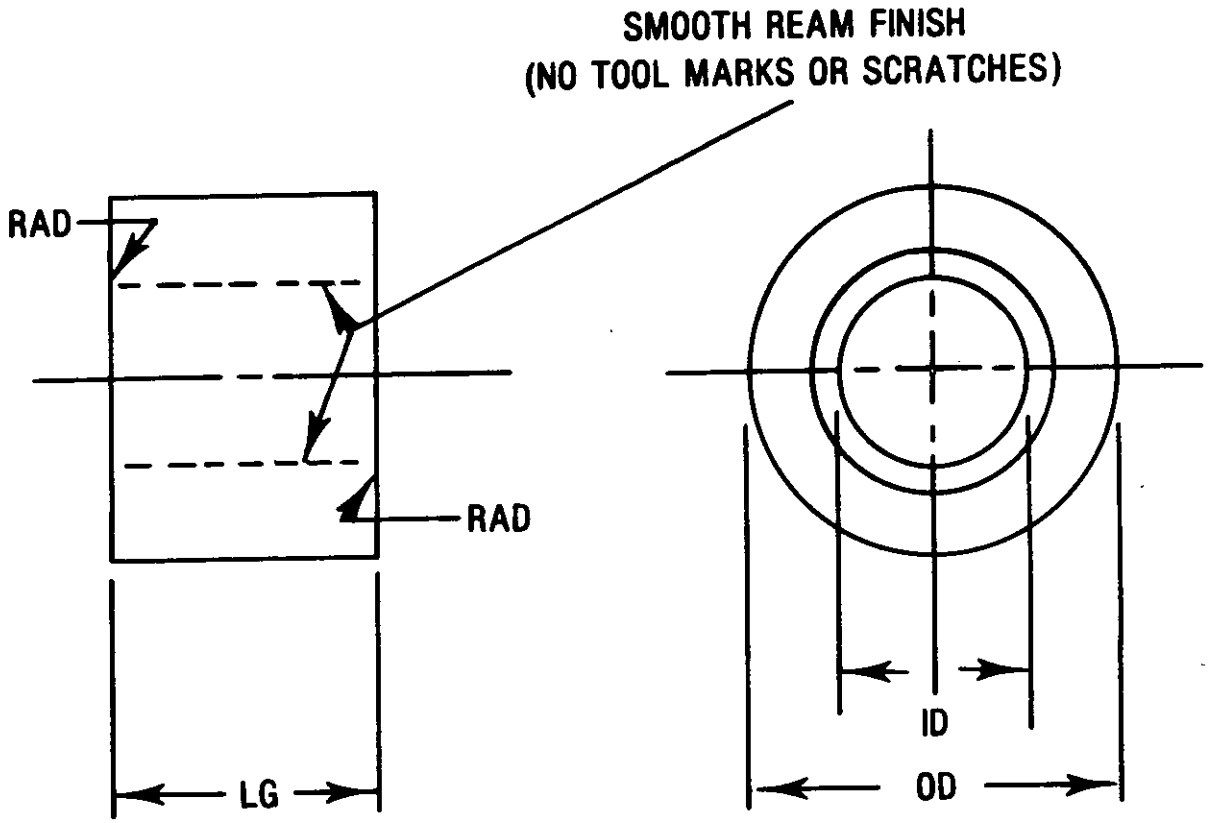


FIGURE 6. Table layout for Frazier air permeability test machine.

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MATERIAL: COMMERCIAL BRASS

Dimensions in inches:

inside diameter (ID)	=	.500 ±.001
outside diameter (OD)		1.0 (approximate)
length (LG)		.750 ±.001
radius (RAD)		.062 (approximate)

FIGURE 7. Bushing.

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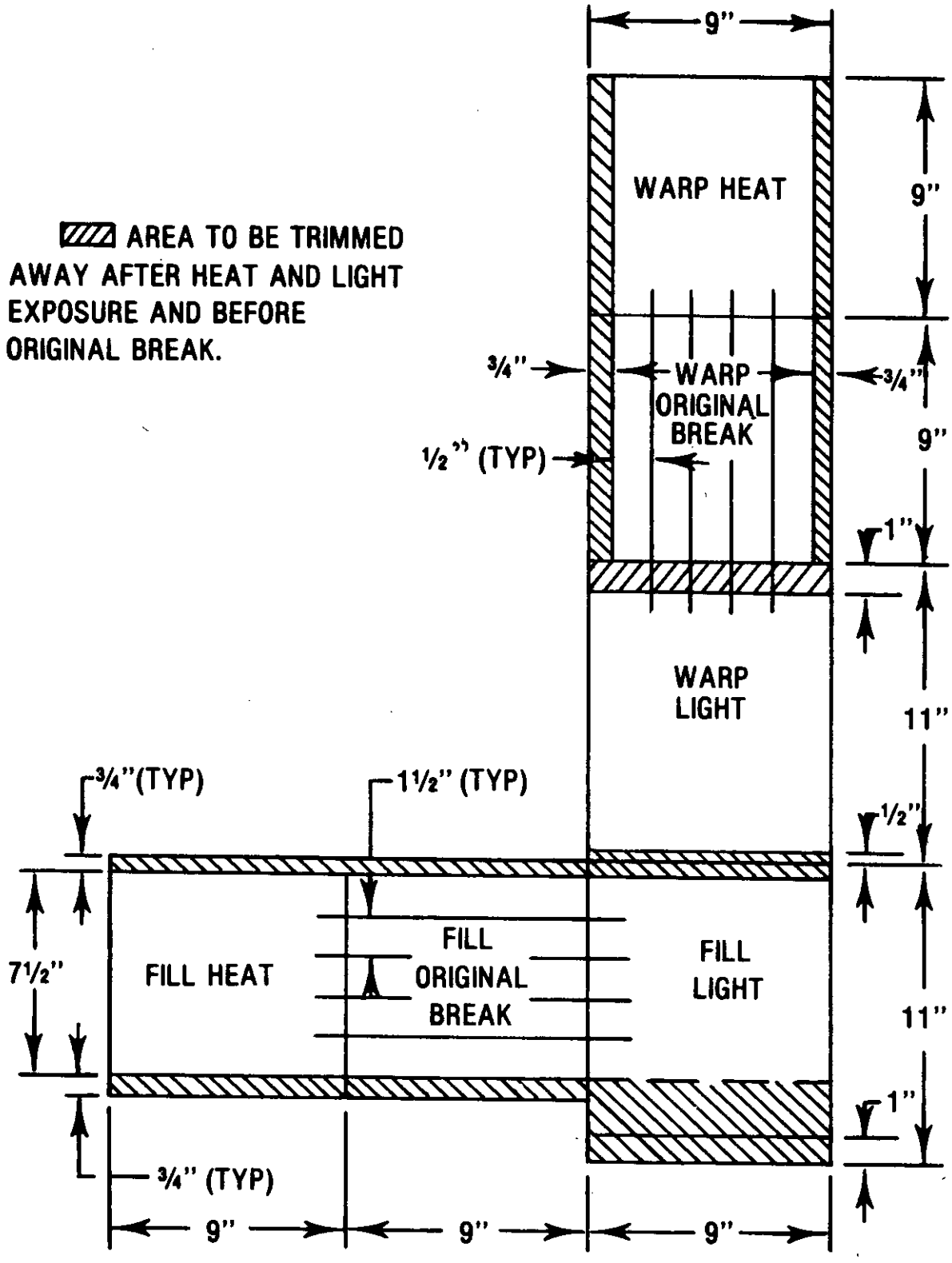


FIGURE 8. Test specimens, original break, heat and light.

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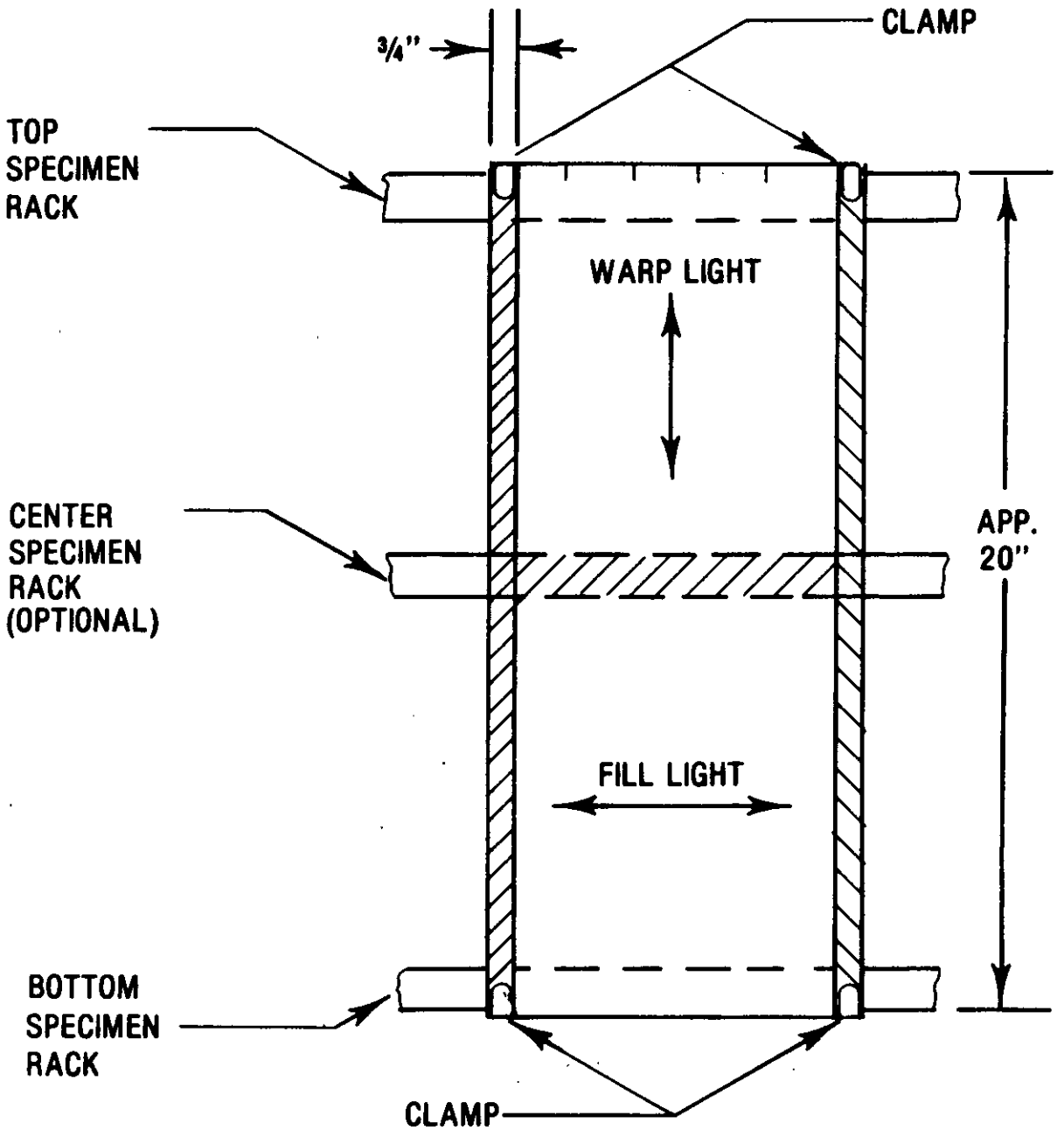


FIGURE 9. Test specimens, resistance to light.

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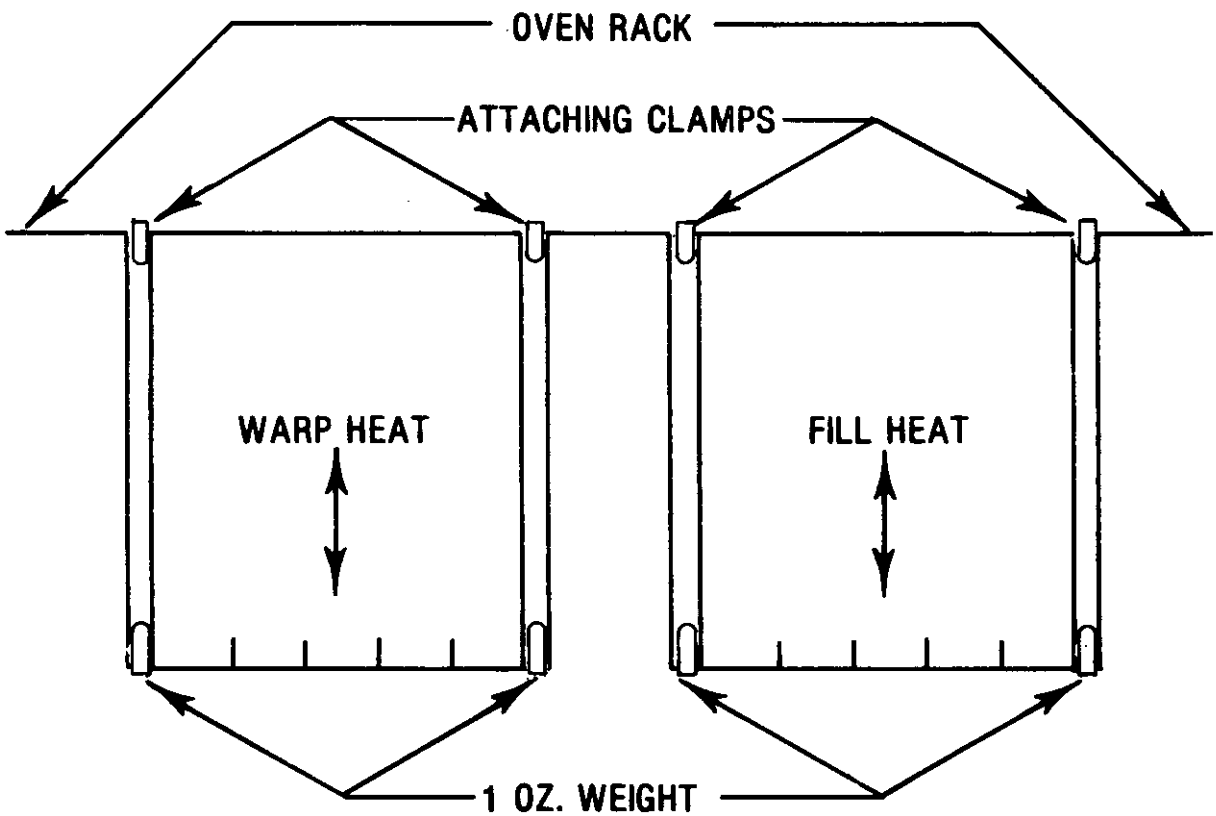


FIGURE 10. Test specimens, resistance to heat.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-87180	Cloth, Parachute, Stretch and Ripstop

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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